How Literacy and Age at Marriage Shape Family Size in Portugal, 1979*

A Statistic Research Using Poisson Regression Models

Shanjie Jiao Xizi Sun

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TBD. IDK. Smash Them

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*Code and data are available at: https://github.com/Jie-jiao05/Family_Size.

1 Introduction

1.1 Why this topic is important

1.2 3 citation relate with our research

1.3 Our objective, goal, how the citation related to, what we gonna do

1.4 Brief introduction of our research

2 Method

2.1 Use GLM, possion, (reason), what is x how to interoperate

3 Result

3.1 Data Description

This dataset originates from the 1979-80 Portuguese Fertility Survey (Instituto Nacional de Estatística (Portugal) 1979), conducted as part of the World Fertility Survey (WFS) program and archived by Princeton University's Office of Population Research (OPR).

The Dataset includes 5,148 observations and 8 variables, covering factors such as literacy, marriage age, pregnancies, children, and regional demographics. Table 1 it shows a summary information of our dataset.

3.2 Model

```
# Load necessary libraries
library(dplyr)
library(MASS)
```

```
Attaching package: 'MASS'

The following object is masked from 'package:patchwork':

    area
```

Table 1: Dataset Summary

```
      X              age         ageMarried         monthsSinceM
 Min.   :   1   Min.   :15.00   Length:5148        Min.   :  0.0
 1st Qu.:1288   1st Qu.:28.75   Class :character   1st Qu.: 73.0
 Median :2574   Median :36.00   Mode  :character   Median :148.0
 Mean   :2574   Mean   :35.30                      Mean   :155.2
 3rd Qu.:3861   3rd Qu.:43.00                      3rd Qu.:230.0
 Max.   :5148   Max.   :49.00                      Max.   :431.0
  pregnancies        children          sons           literacy
 Min.   : 0.000   Min.   : 0.00   Min.   : 0.000   Min.   :0.0000
 1st Qu.: 1.000   1st Qu.: 1.00   1st Qu.: 0.000   1st Qu.:1.0000
 Median : 2.000   Median : 2.00   Median : 1.000   Median :1.0000
 Mean   : 2.282   Mean   : 2.26   Mean   : 1.187   Mean   :0.8871
 3rd Qu.: 3.000   3rd Qu.: 3.00   3rd Qu.: 2.000   3rd Qu.:1.0000
 Max.   :16.000   Max.   :17.00   Max.   :10.000   Max.   :1.0000
  region10.20k       region20k.      regionlisbon     regionlt10k
 Min.   :0.00000   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000
 1st Qu.:0.00000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000
 Median :0.00000   Median :0.0000   Median :0.0000   Median :1.0000
 Mean   :0.08411   Mean   :0.1132   Mean   :0.0913   Mean   :0.6803
 3rd Qu.:0.00000   3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:1.0000
 Max.   :1.00000   Max.   :1.0000   Max.   :1.0000   Max.   :1.0000
  regionporto
 Min.   :0.00000
 1st Qu.:0.00000
 Median :0.00000
 Mean   :0.03108
 3rd Qu.:0.00000
 Max.   :1.00000
```

```
The following object is masked from 'package:plotly':

    select


The following object is masked from 'package:dplyr':

    select
```

```r
# Read the cleaned dataset
data <- read.csv("~/Family_Size/Data/cleaned_portugal.csv")

# Ensure monthsSinceM is positive to prevent log(0) errors
data <- data %>%
  filter(monthsSinceM > 0) %>%
  mutate(log_monthsSinceM = log(monthsSinceM))  # Compute log offset

# Fit Poisson regression model with offset
poisson_model <- glm(children ~ ageMarried + literacy,
                     family = poisson(link = "log"),
                     offset = log_monthsSinceM,
                     data = data)

# Display model summary
summary(poisson_model)
```

```
Call:
glm(formula = children ~ ageMarried + literacy, family = poisson(link = "log"),
    data = data, offset = log_monthsSinceM)

Coefficients:
                  Estimate Std. Error z value Pr(>|z|)
(Intercept)       -4.07811    0.07882 -51.741  < 2e-16 ***
ageMarried15to18   0.02634    0.08267   0.319    0.750
ageMarried18to20   0.01214    0.08036   0.151    0.880
ageMarried20to22  -0.02037    0.07993  -0.255    0.799
ageMarried22to25  -0.04959    0.07952  -0.624    0.533
ageMarried25to30  -0.03615    0.08072  -0.448    0.654
ageMarried30toInf -0.02792    0.09558  -0.292    0.770
literacy          -0.15942    0.02355  -6.770 1.29e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 5654.3  on 5145  degrees of freedom
Residual deviance: 5600.1  on 5138  degrees of freedom
AIC: 18047

Number of Fisher Scoring iterations: 5
```

3.3 Model selection (use model with offset)

3.4 Use likelyhood rario test to do the performance , to see the efficiency between model

3.5 Test overdispersion

4 Conclusion

Appendix

References

Instituto Nacional de Estatística (Portugal). 1979. *Portugal World Fertility Survey (WFS), 1979-80.* World Fertility Survey (WFS), Demographic; Health Surveys (DHS) Program. https://wfs.dhsprogram.com/index.cfm?ccode=pt.